

CASE STUDY · POWER BI CAPSTONE PROJECT

Esports Market Intelligence

Cleaning, modeling and visualizing a \$879.4M market to guide an investment decision

Prepared by

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Role

Data Analyst

Tools

Power BI, Power Query, DAX

Dataset

ESport_Earnings.csv — 504 records, 503 titles

1. Project Overview

A sports entertainment and investment firm was evaluating market entry into competitive gaming (esports). As the data analyst on the engagement, I was asked to clean, model and visualize a market dataset in Power BI and deliver an interactive report the firm's leadership could use to guide capital allocation.

The finished report is a two-page Power BI dashboard covering market size and genre concentration, and geographic dependency and tournament efficiency, backed by a dedicated DAX measures table and a hover tooltip with per-title averages.

\$879.4M Total prize money tracked	503 Unique game titles	40,696 Tournaments tracked	11 Genres represented
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2. Business Problem

Leadership's decision hinged on three questions that shaped every visual in the dashboard:

- **Where is the money?** Which genres and titles concentrate the market's prize pools.
- **Which games use money efficiently?** Prize money per player and per tournament, not just totals.
- **Which titles depend on a single country?** Regional concentration relevant to sponsorship and geo-targeting.

3. Data & Methodology

Pipeline: CSV → Power Query → Data model (fact table + dedicated *_Measures* table) → DAX → Dashboard.

Cleaning decisions

Issue	Finding	Resolution
Missing geography	48 of 504 rows had a null Top_Country.	Replaced with "Unknown" so no rows are excluded when a slicer or filter is applied to that field.
Year typo	Forza Motorsport 4 was recorded with release year "11".	Corrected to 2011 with a conditional transform in Power Query.

Issue	Finding	Resolution
Duplicate title	Battalion 1944 appears twice; one row is entirely empty (IdNo 537).	Row kept in the 504-record table; Games Tracked uses DISTINCTCOUNT, so it is counted once (503 titles) and the empty row adds zero to every total.
Data typing	Money and count fields were untyped on import.	TotalMoney and Top_Country_Earnings set to fixed decimal (currency); PlayerNo and TournamentNo set to whole number.

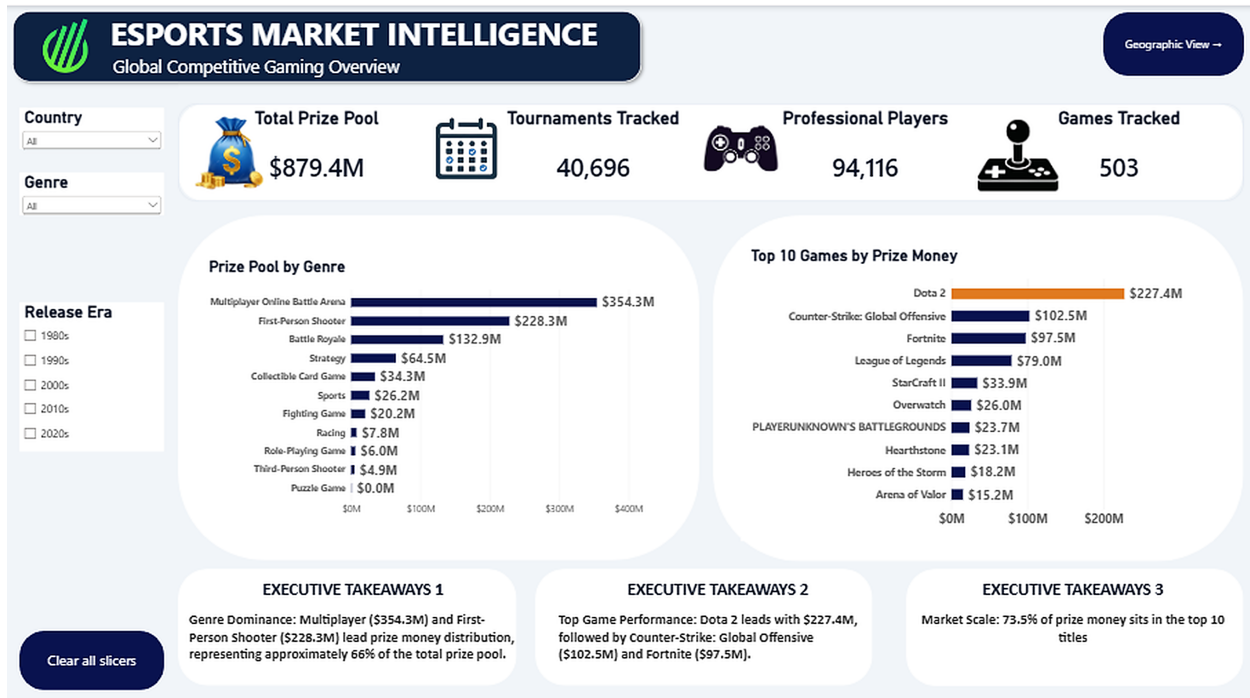
Core DAX measures

Measure	DAX
Total Prize Money	SUM(ESport_Earnings[TotalMoney])
Total Tournaments	SUM(ESport_Earnings[TournamentNo])
Total Players	SUM(ESport_Earnings[PlayerNo])
Avg Prize Per Tournament	DIVIDE([Total Prize Money],[Total Tournaments],0)
Avg Prize Per Player	DIVIDE([Total Prize Money],[Total Players],0)
Top Country Prize Share %	DIVIDE(SUM(Top_Country_Earnings),[Total Prize Money],0)

DIVIDE() was used throughout instead of the / operator so that titles with zero recorded players or tournaments return 0 rather than an error.

4. Dashboard — Executive Overview

The first page frames market size and shows where prize money concentrates by genre and by title, with slicers for Country, Genre and Release Era.



Executive Overview page, built in Power BI. Source: ESport_Earnings.csv, 503 titles after cleaning.

73.5%

of all prize money sits in the top 10 titles

66%

of prize money is MOBA + First-Person Shooter combined

\$227.4M

won by Dota 2, the single largest title

5. Dashboard — Geographic Reach & Tournament Efficiency

The second page examines which countries lead each title's earnings and plots tournament activity against prize money on a log scale, so both very small and very large titles are visible on one chart.



GEOGRAPHIC REACH & TOURNAMENT EFFICIENCY

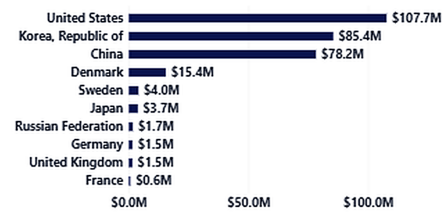
Global prize concentration, tournament activity and player participation

Executive
Overview

Regional Prize Concentration

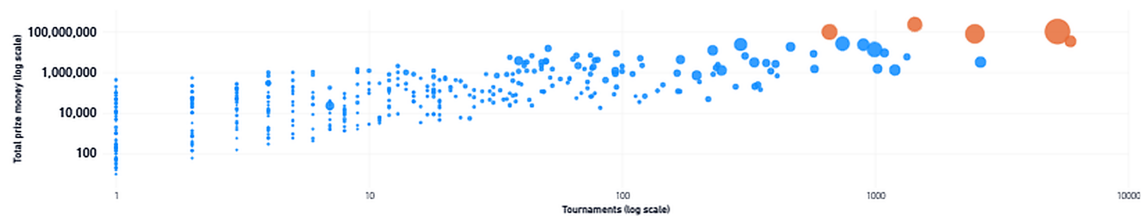
GameName	Top_Country	Top_Country Prize Share %	Sum of Top_Country_Earnings
Dota 2	China	28.6%	\$65,060,611.5
Fortnite	United States	34.3%	\$33,422,756.4
League of Legends	Korea, Republic of	33.8%	\$26,739,426.1
StarCraft II	Korea, Republic of	60.8%	\$20,635,189.1
Counter-Strike: Global Offensive	Denmark	15.0%	\$15,379,018.3
Total		33.5%	\$280,788,792.3

Prize Earnings of Each Title's Top Country



Tournament Activity vs Prize Money

Orange = top 5 titles by prize money: Dota 2, Counter-Strike: GO, Fortnite, League of Legends, StarCraft II



Geographic Reach & Tournament Efficiency page, built in Power BI. Top_Country_Earnings records each title's leading country only, not full national totals.

\$107.7M

US top-country earnings, the highest of any country

93.5%

of Halo 2 prize money went to US players

91.8%

of StarCraft: Brood War prize money went to South Korea

6. Key Findings

Market concentration. The top 10 titles hold 73.5% of the \$879.4M tracked prize pool — the market rewards depth, not breadth.

Genre dominance. MOBA alone is about 40% of prize money from just 22 of 503 titles; MOBA and First-Person Shooter together are roughly 66%.

A crowded, low-yield genre. Fighting games are the most numerous genre (160 titles) but earn only about 2.3% of total prize money.

Efficiency gap. MOBA pays roughly \$22.9K in prize money per player versus about \$1.6K for Fighting games — a 14x difference in payout efficiency.

Regional dependency. Some flagship titles are concentrated in one country: Halo 2 is about 93.5% US, and StarCraft: Brood War is about 91.8% South Korea.

7. Strategic Recommendations

1. Anchor in MOBA and First-Person Shooter

These two genres represent about 66% of tracked prize money, with MOBA paying the highest amount per player. Recommended action: flagship-title and league-level sponsorships.

2. Test Battle Royale as an emerging growth genre

Only 11 titles make up this genre, yet it holds about 15% of prize money and roughly \$110K per tournament, around five times the market average of \$21.6K. Recommended action: a small, staged sponsorship pilot before committing further.

3. Use regional concentration for targeted deals

Titles like Halo 2 and StarCraft: Brood War are heavily tied to one country, which suits geo-targeted brand deals; low-yield, high-title-count genres such as Fighting games are better suited to community-level spend rather than flagship investment.

8. Conclusion

The esports prize-money market is highly concentrated rather than evenly distributed: a small number of titles, genres and countries account for a disproportionate share of the money in play. For an investor, this argues for anchoring capital in proven, high-payout genres and titles, using data to test promising but smaller genres such as Battle Royale, and treating the long tail of low-yield, high-title-count genres as community engagement rather than flagship investment.

Methodology notes and limitations

- “Professional Players” is a sum of PlayerNo per title, so a person competing in two titles is counted twice; it should be read as player entries, not unique individuals.

- Top_Country and Top_Country_Earnings record only each title's single leading country, so the country chart shows dependency on one country, not that country's total esports earnings.
- Titles with zero recorded prize money, players or tournaments were kept in the dataset rather than removed, since they are real, valid records; DIVIDE() protects every ratio measure from returning an error.

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